

A common framework to study the evolution and development of human communication

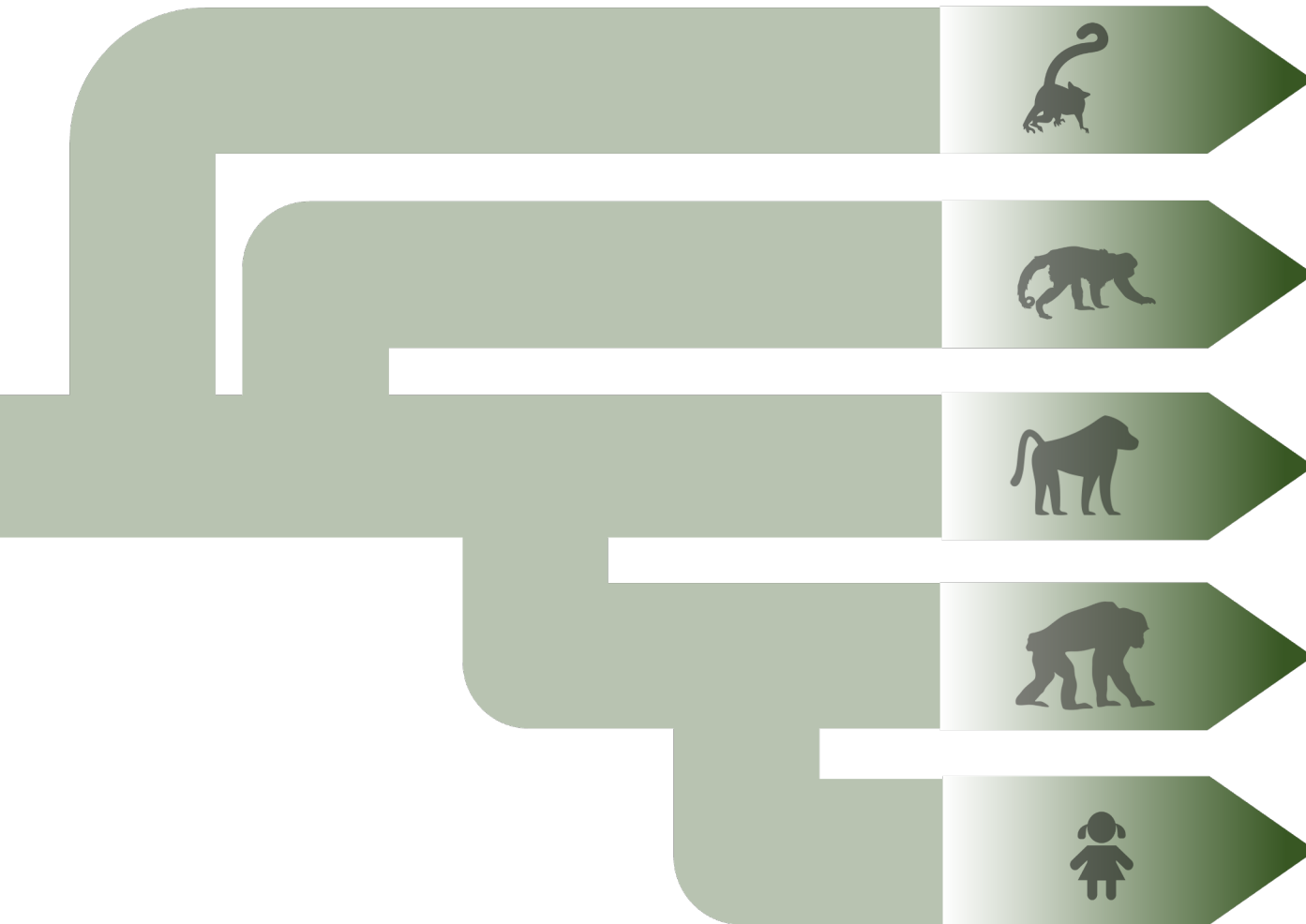
Manuel Bohn

`manuel.bohn@leuphana.de // https://manuelbohn.github.io // @elmanubohn.bsky.social`

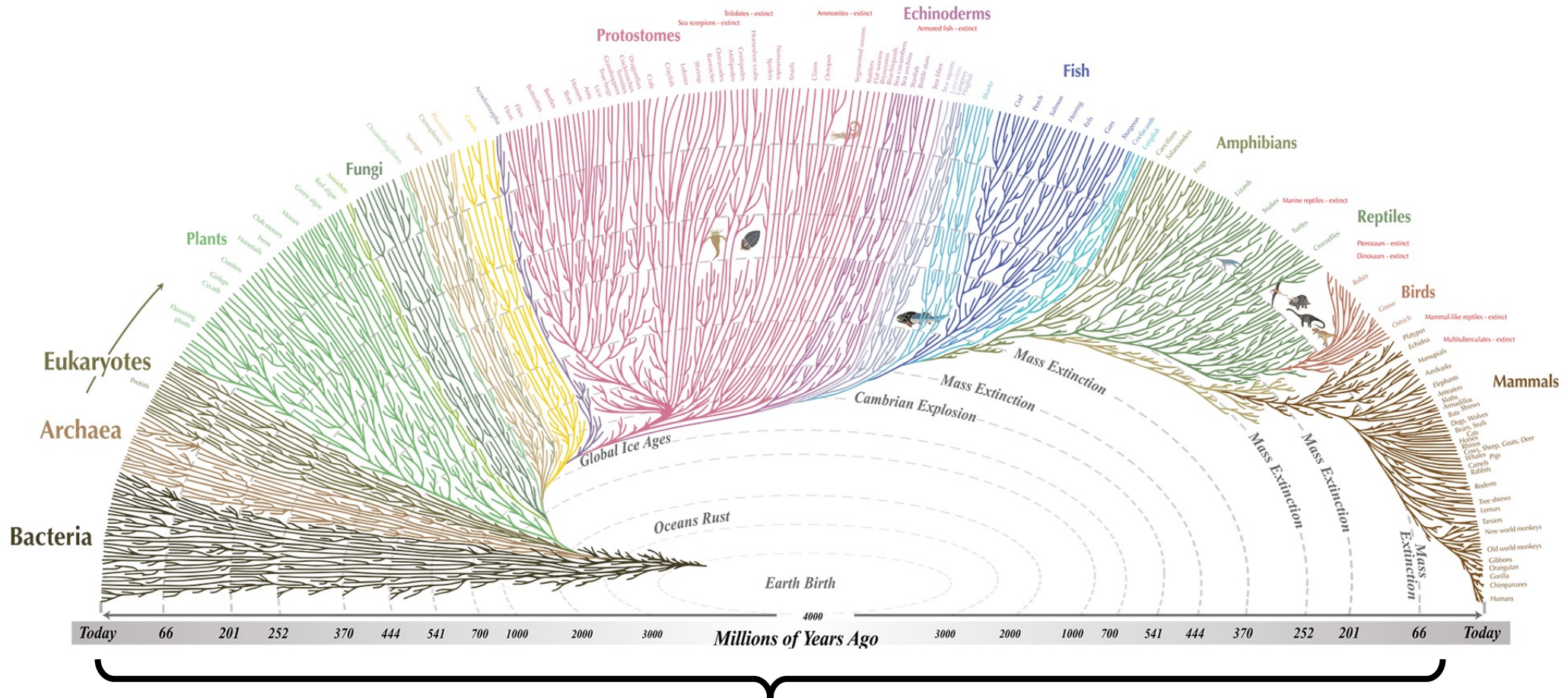


LEUPHANA
UNIVERSITÄT LÜNEBURG

An evolutionary perspective on human communication

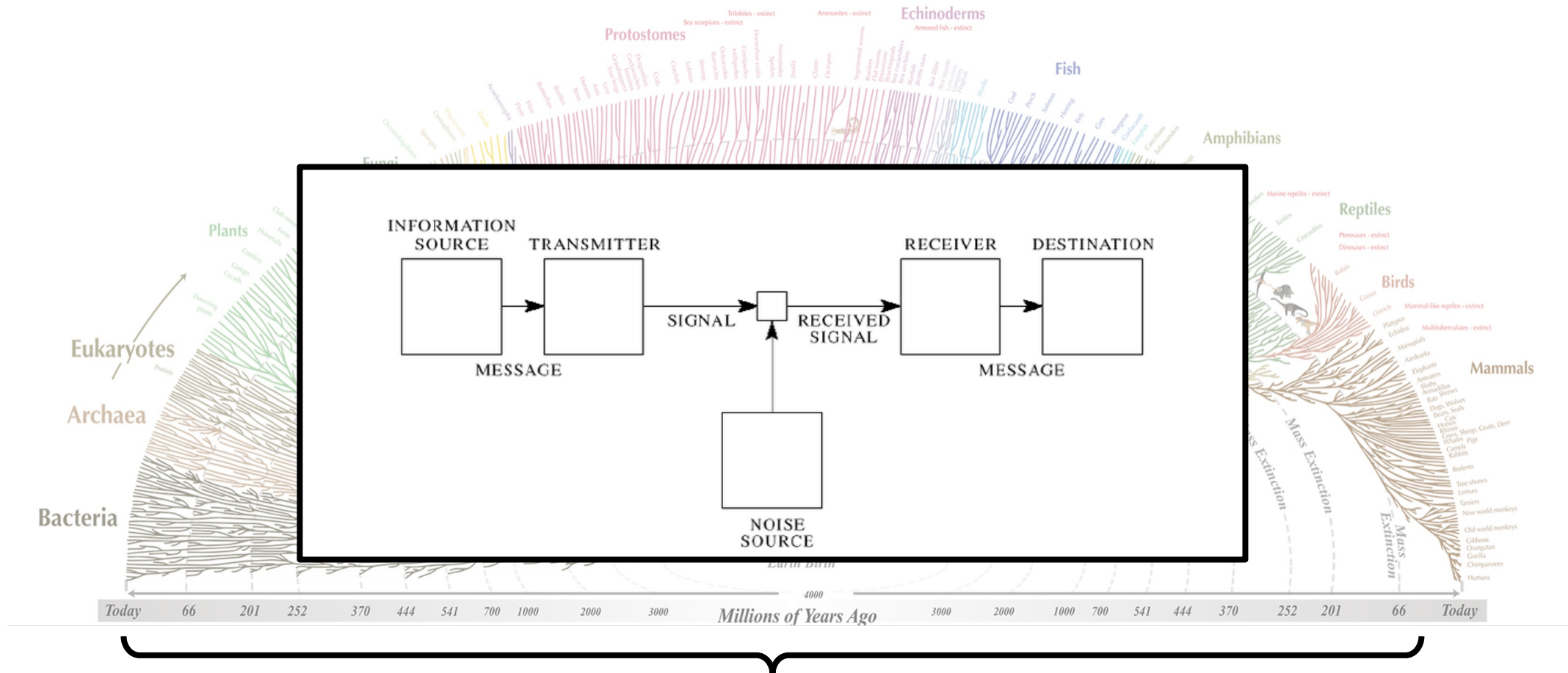


Models of communication



Code-based communication (natural meaning)

Models of communication



Code-based communication (natural meaning)

Models of communication



Models of communication



Ostensive-inferential communication (non-natural meaning)

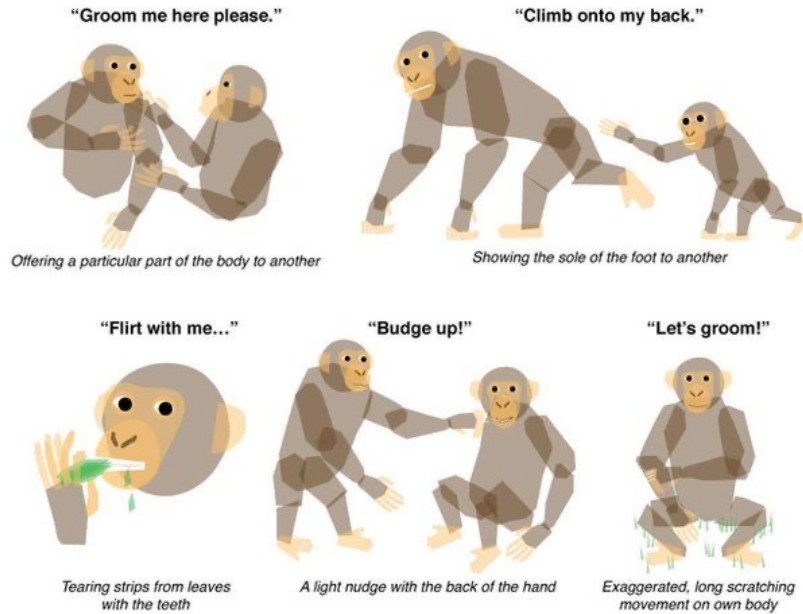
Models of communication



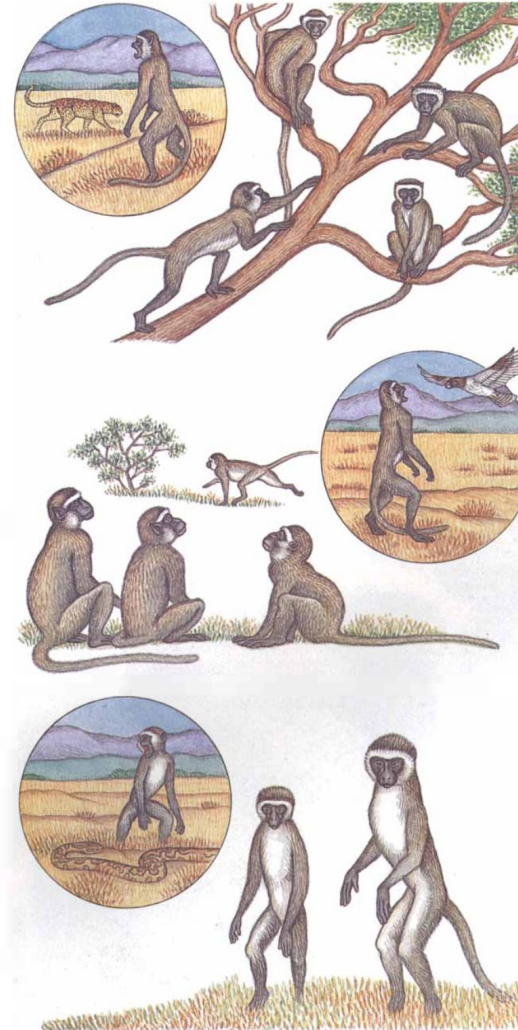
?

Ostensive-inferential communication (non-natural meaning)

'Meanings' of primate signals



Source: Hobaiter and Byrne: The Meanings of Chimpanzee Gestures, Current Biology (2014)

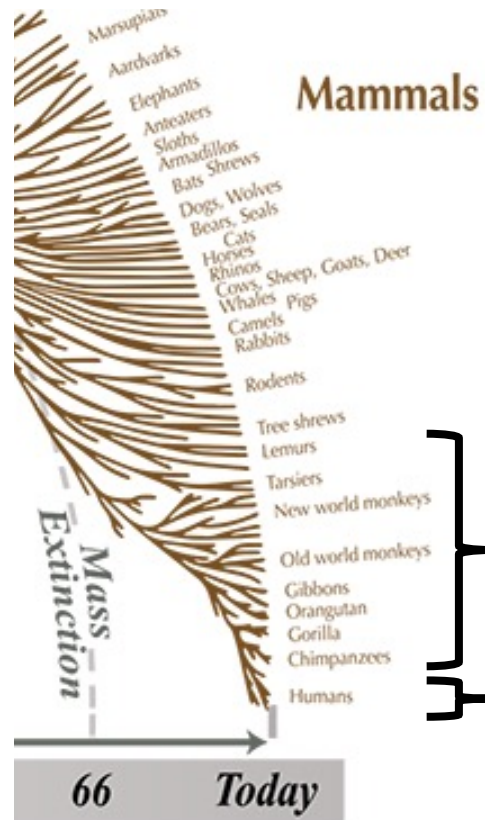


Chimpanzee		Equivalent human facial expression?	
Facial expression	Prototype AUs	Facial expression	Prototype AUs
	Bulging-lip display AU17 AU24		Angry face AU4 AU5 AU7 AU17 AU24
	Silent bared-teeth display AU10 AU12 AU16 AU25		Smile AU6 AU12 AU25
	Scream AU10 AU12 AU16 AU25 AU27		Scream AU4 AU6 AU7 AU10 AU12 AU16 AU25 AU27
	Pant-hoot AU22 AU25 AU26		?
	Relaxed open mouth AU12 AU25 AU26		Laughter AU6 AU12 AU25 AU26

Note: From "The Evolution of Human Emotion" by J. A. Parr & R. M. Waller in *Evolution of Neuroscience*

Graham, Hobaiter, Ounsley, Furuichi & Byrne, 2018
Hobaiter & Byrne, 2014
Seyfarth & Cheney, 1992
Parr, Waller, Vick & Bard, 2007

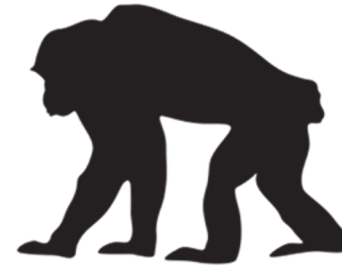
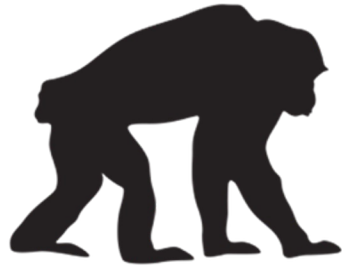
Models of communication

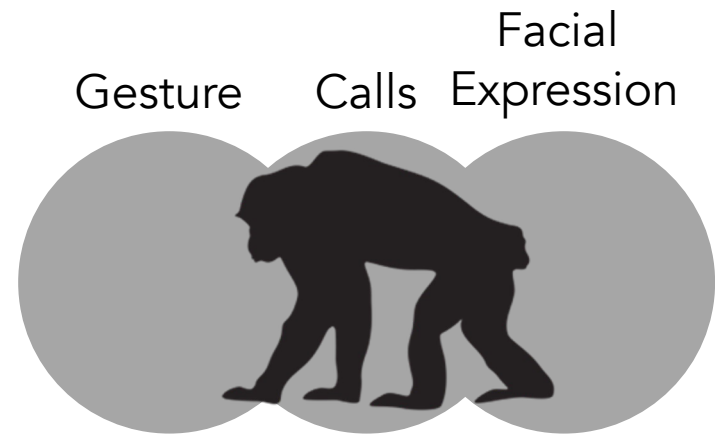
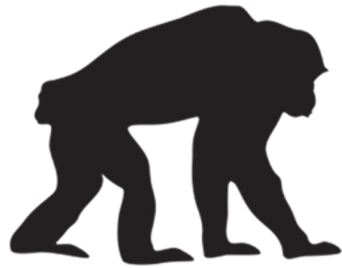


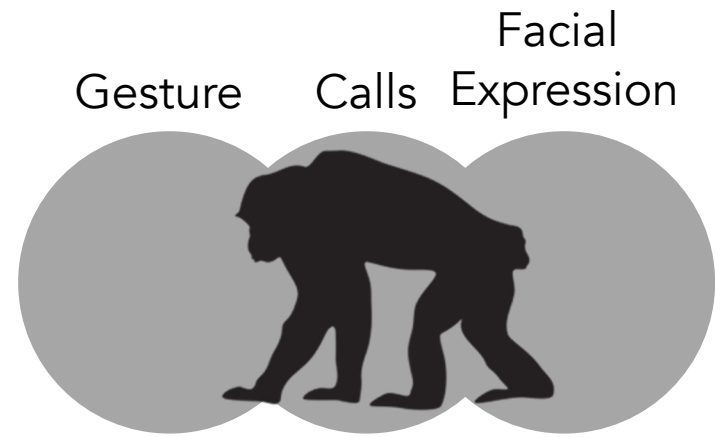
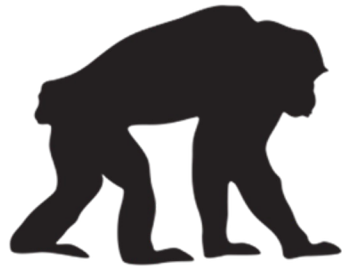
Social-inferential (intentional) communication?

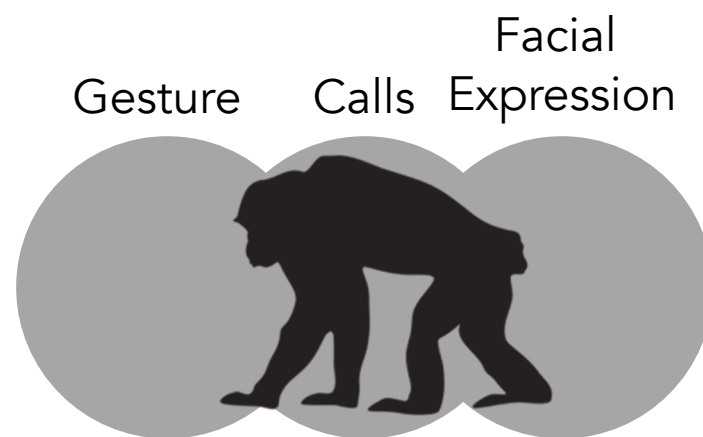
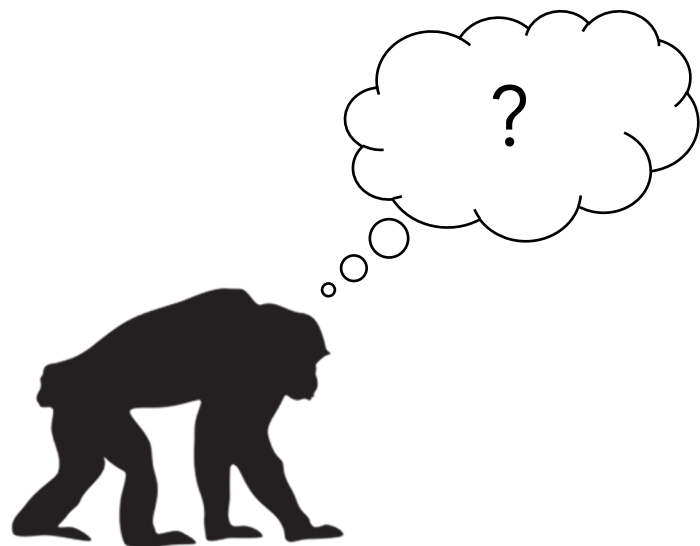
Ostensive-inferential communication (non-natural meaning)

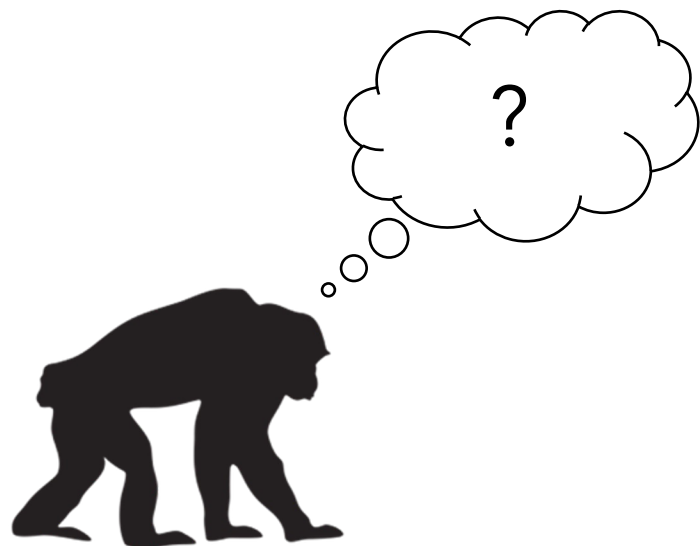
A model of primate communication?



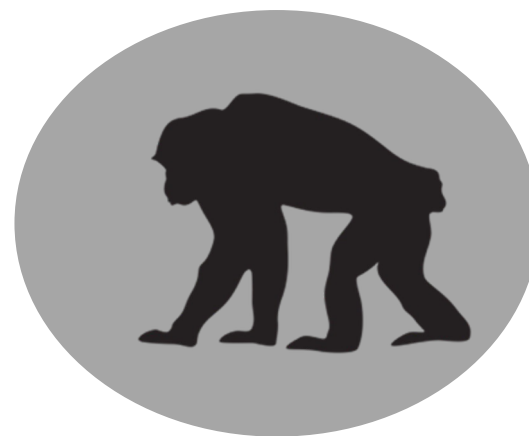




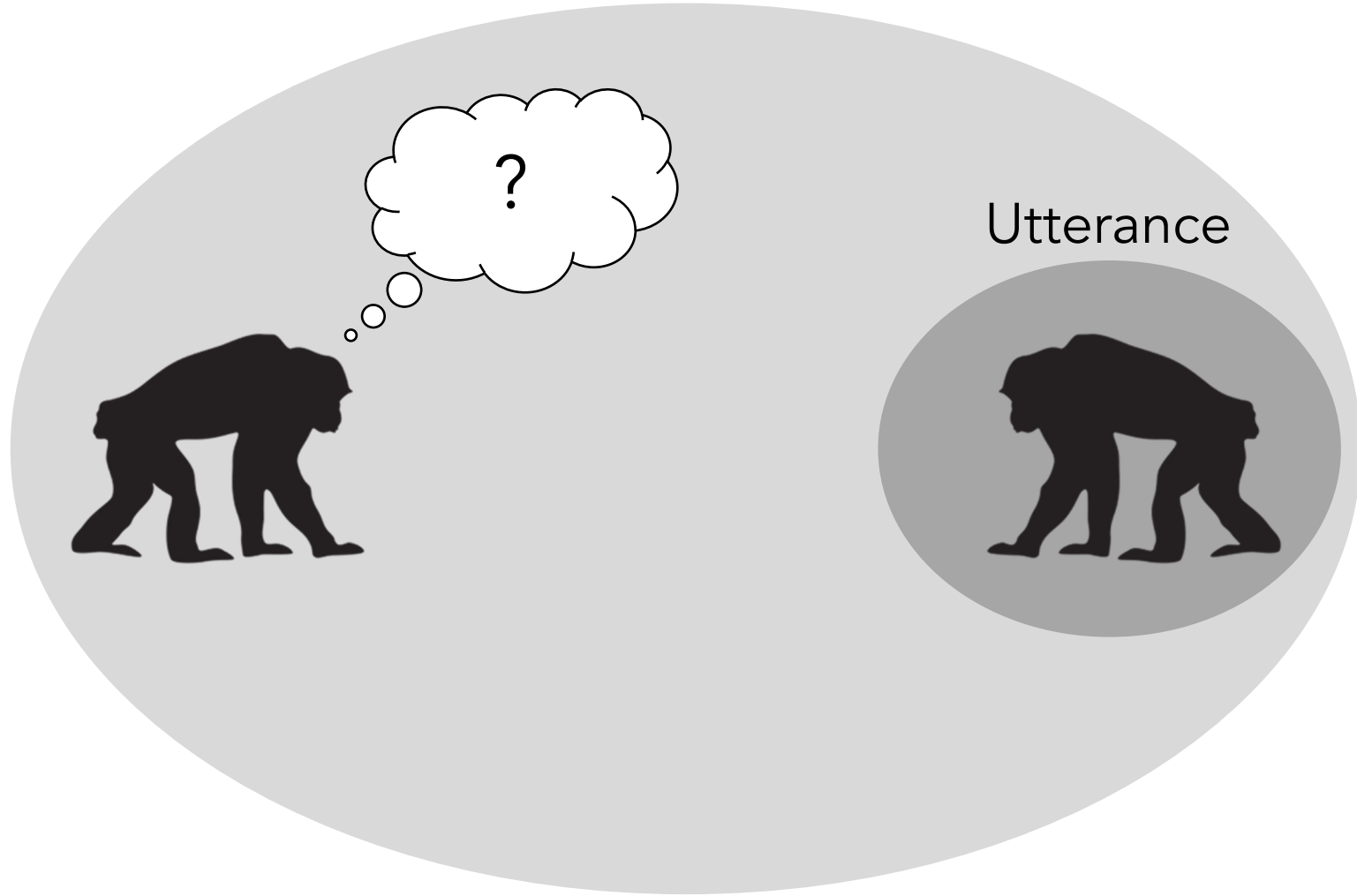




Utterance

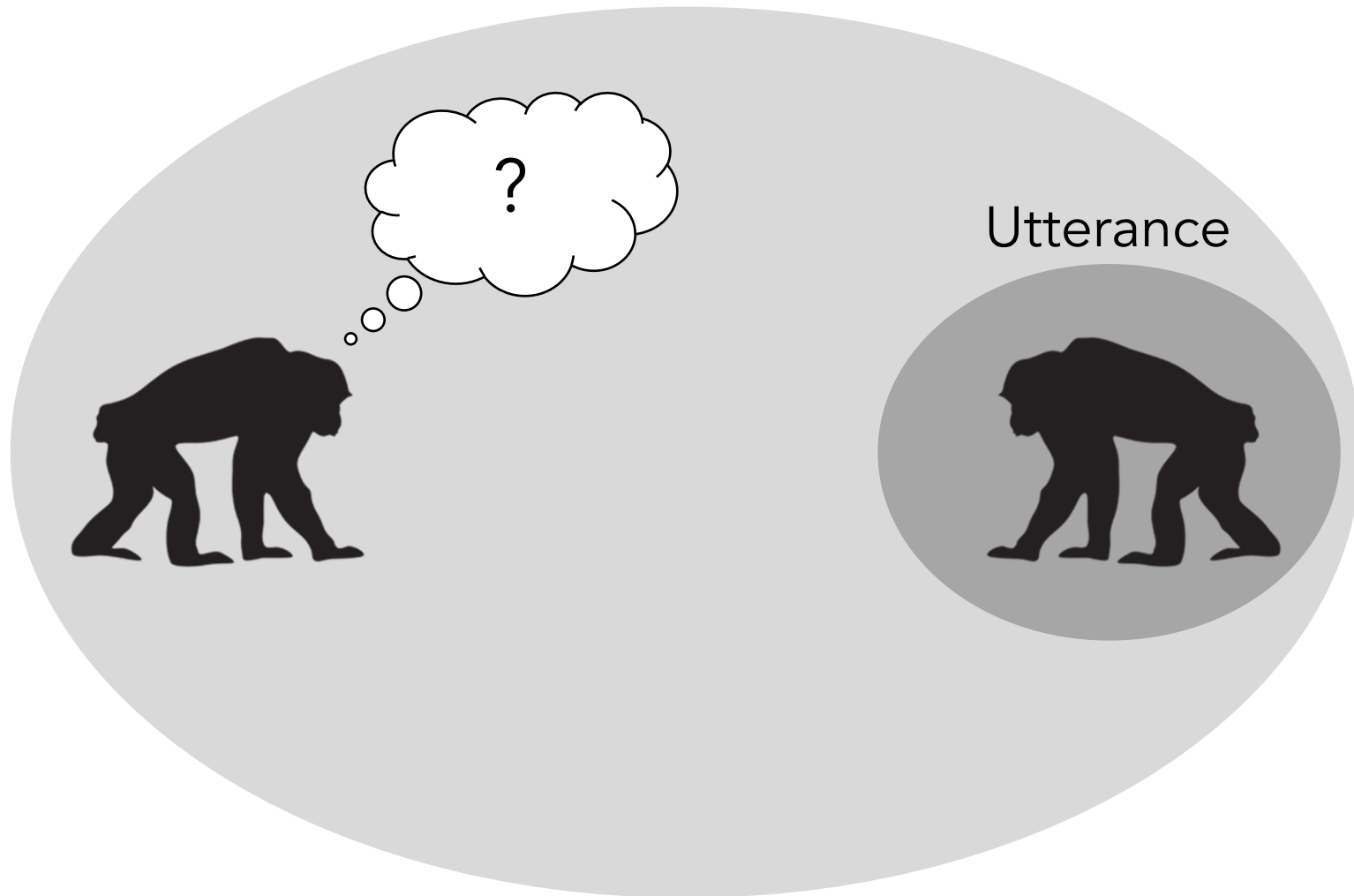


Social context



Utterance

Social context



$$P_R(i|u) \propto P_S(u|i)P(i)$$

Utterance

Semantics of components

$$P_S(u|i) = P(g, f, \dots | i) = \mathcal{L}(g, i | \theta_g) \mathcal{L}(f, i | \theta_f) \dots$$

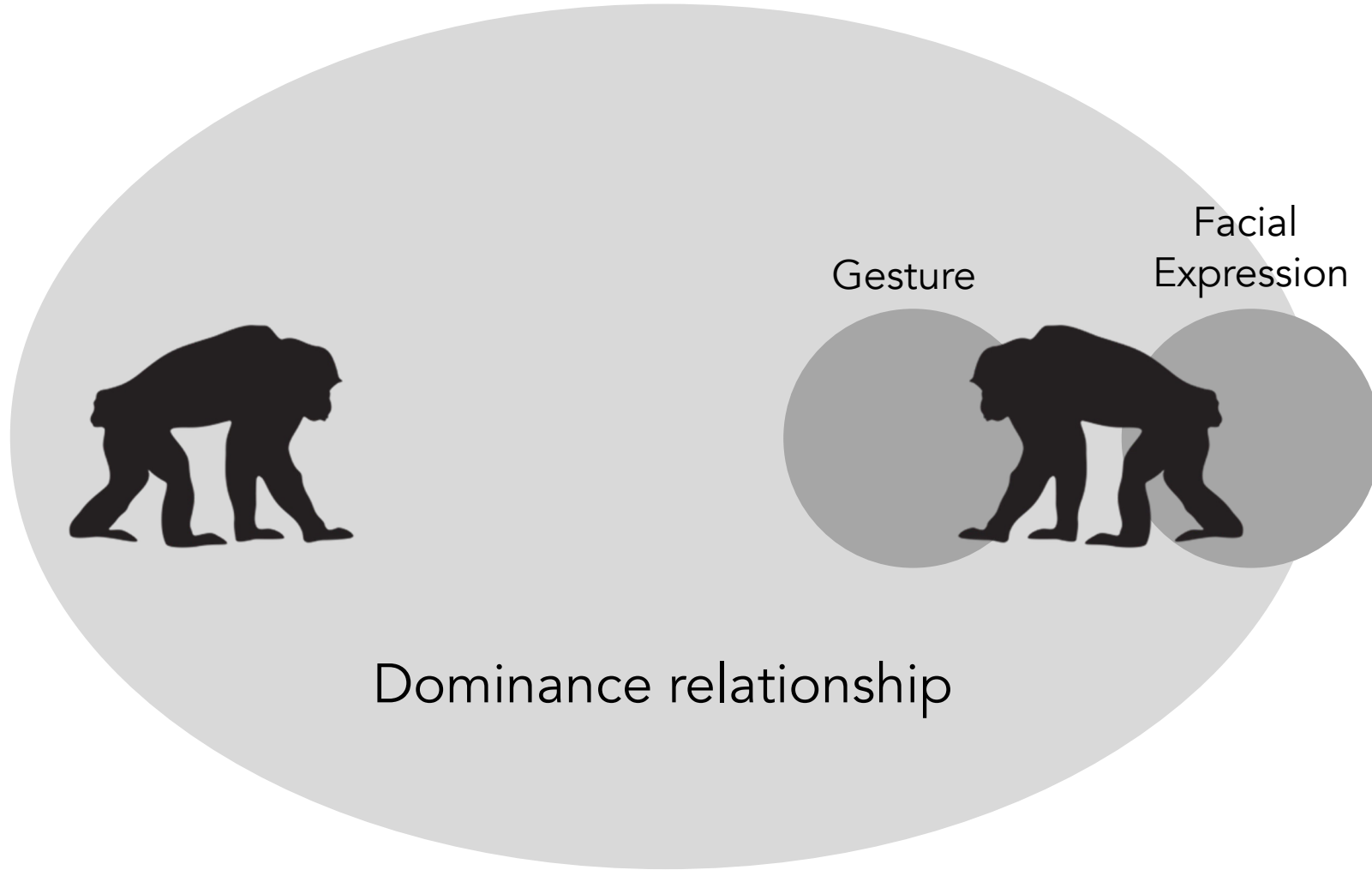
$$P_R(i|u) \propto P_S(u|i) P(i)$$

$$P(i) = P(i | c, s, \dots) = \rho_c \rho_s \dots$$

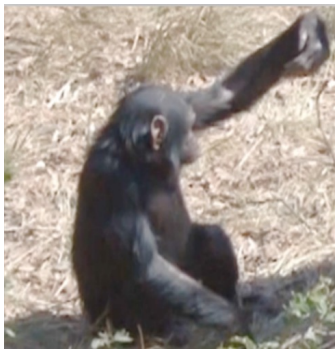
Social context

Strength of components

Social context

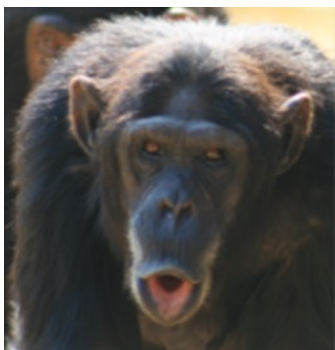
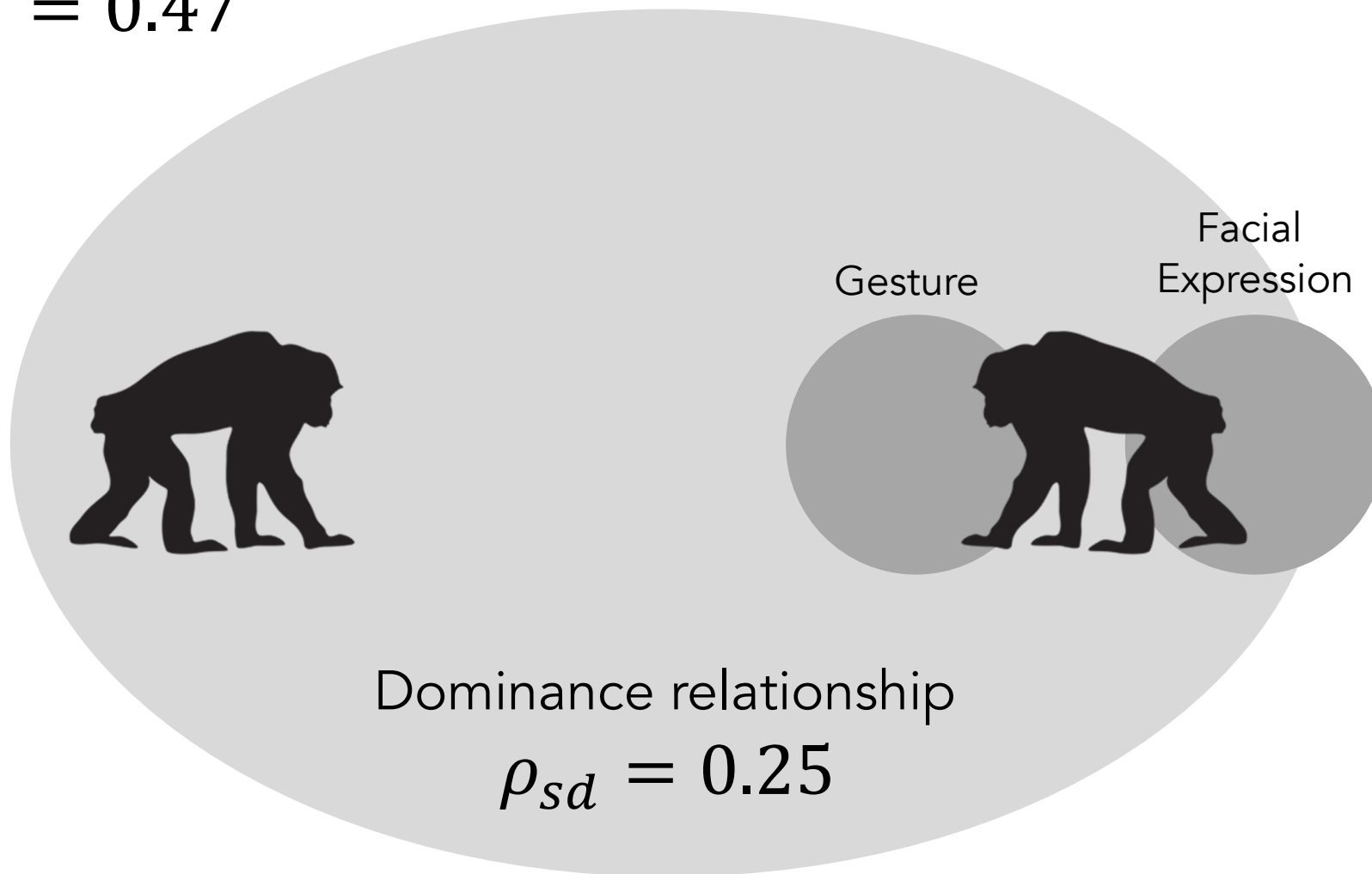


$$P_R(i|u) \propto P_S(u|i)P(i)$$



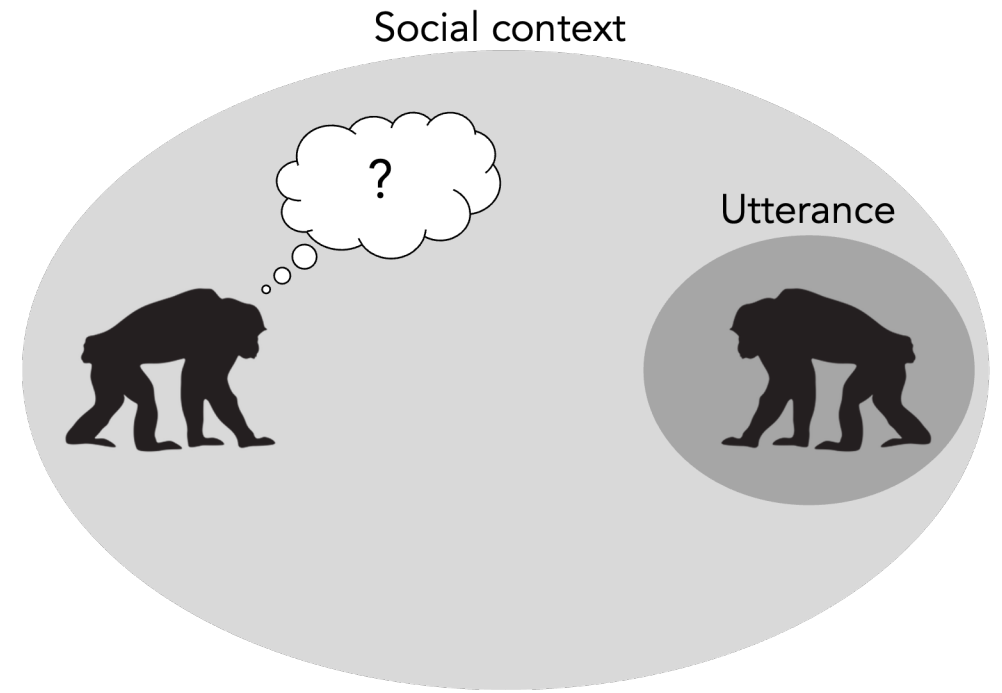
$$\theta_{gb} = 0.47$$

Social context

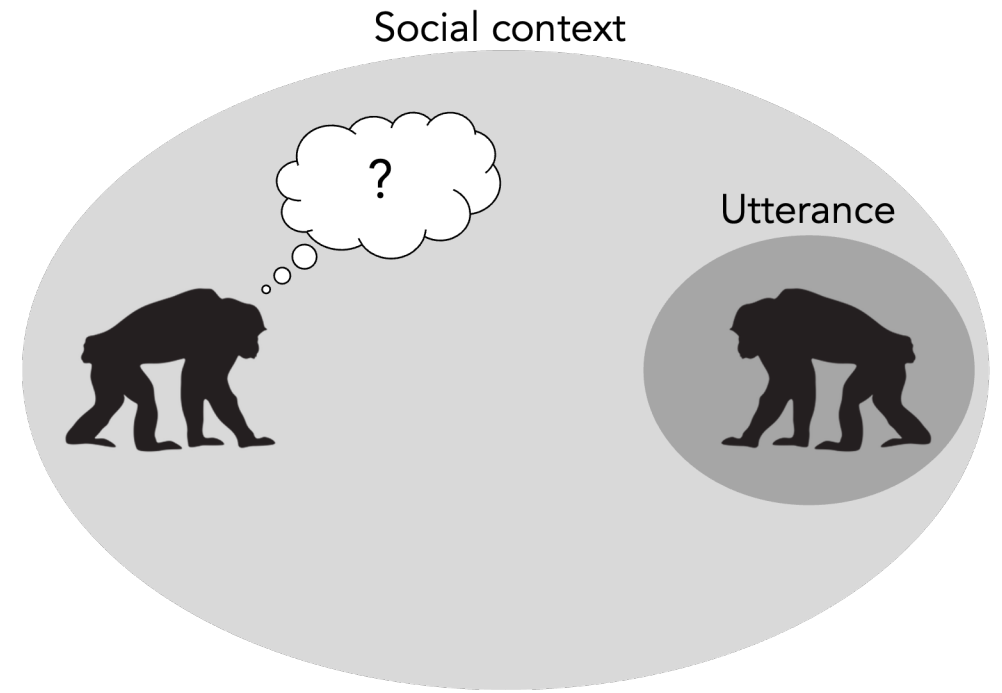


$$\theta_{ff} = 0.9$$

$$P_R(i|u) \propto P_S(u|i)P(i)$$



- Which components are necessary (for different species)?
- How are they related to one another?
- Which components are most important?
- What are the semantics of individual signals?
- What changed in the course of human evolution?



- Which components are necessary (for different species)?
- How are they related to one another?
- Which components are most important?
- What are the semantics of individual signals?
- What changed in the course of human evolution?

Next steps: inferring semantics



Apparently
Satisfactory
Outcomes

Lexical Association Model

$$P_R(a_k | m_j, L, U, \alpha) \propto \exp(\alpha EU(a_k, m_j, L, U))$$

Shared lexicon

No recursive reasoning

Inferring semantics



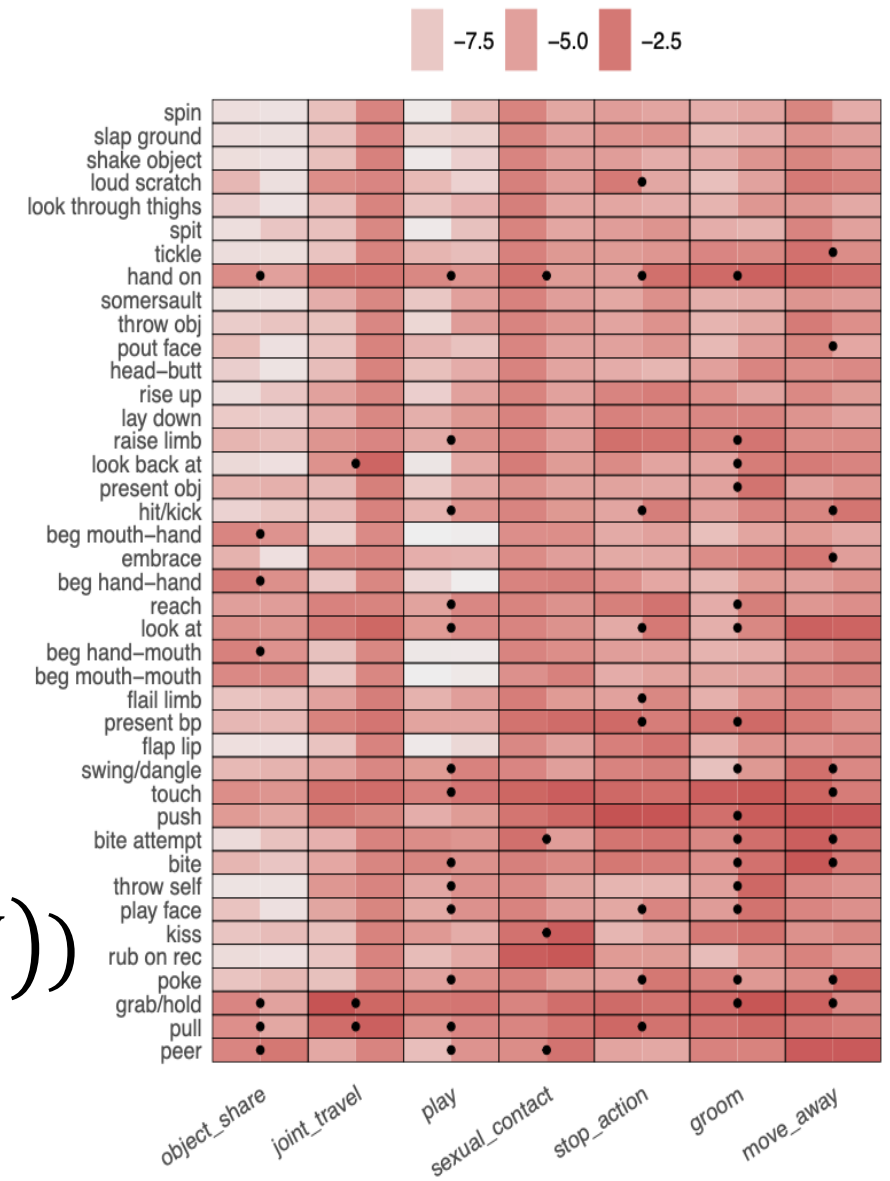
Apparently
Satisfactory
Outcomes

Lexical Association Model

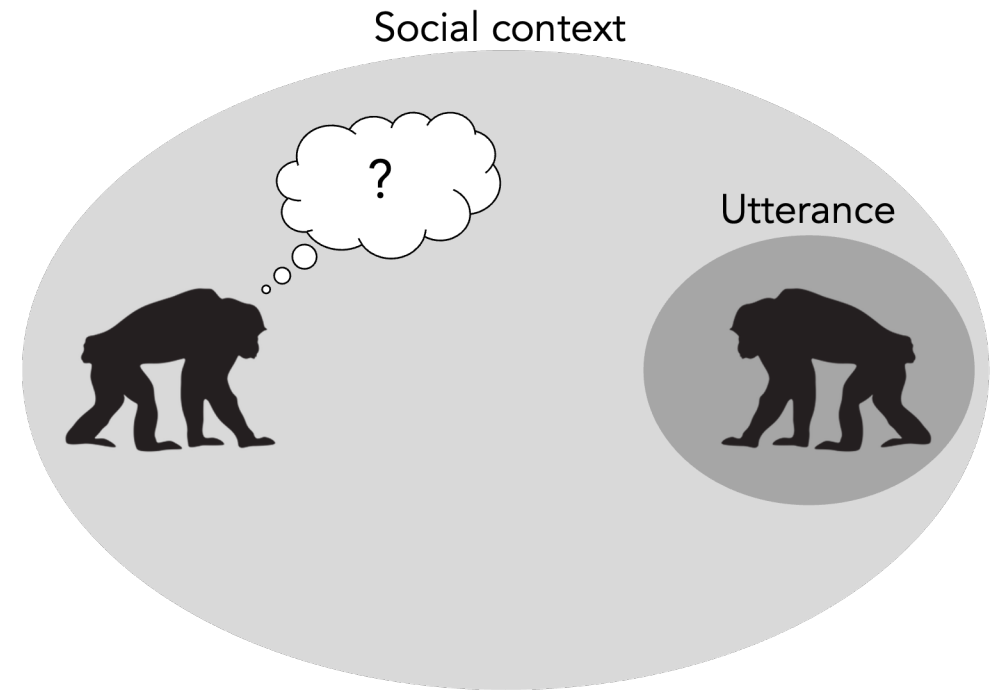
$$P_R(a_k | m_j, L, U, \alpha) \propto \exp(\alpha EU(a_k, m_j, L, U))$$

Shared lexicon

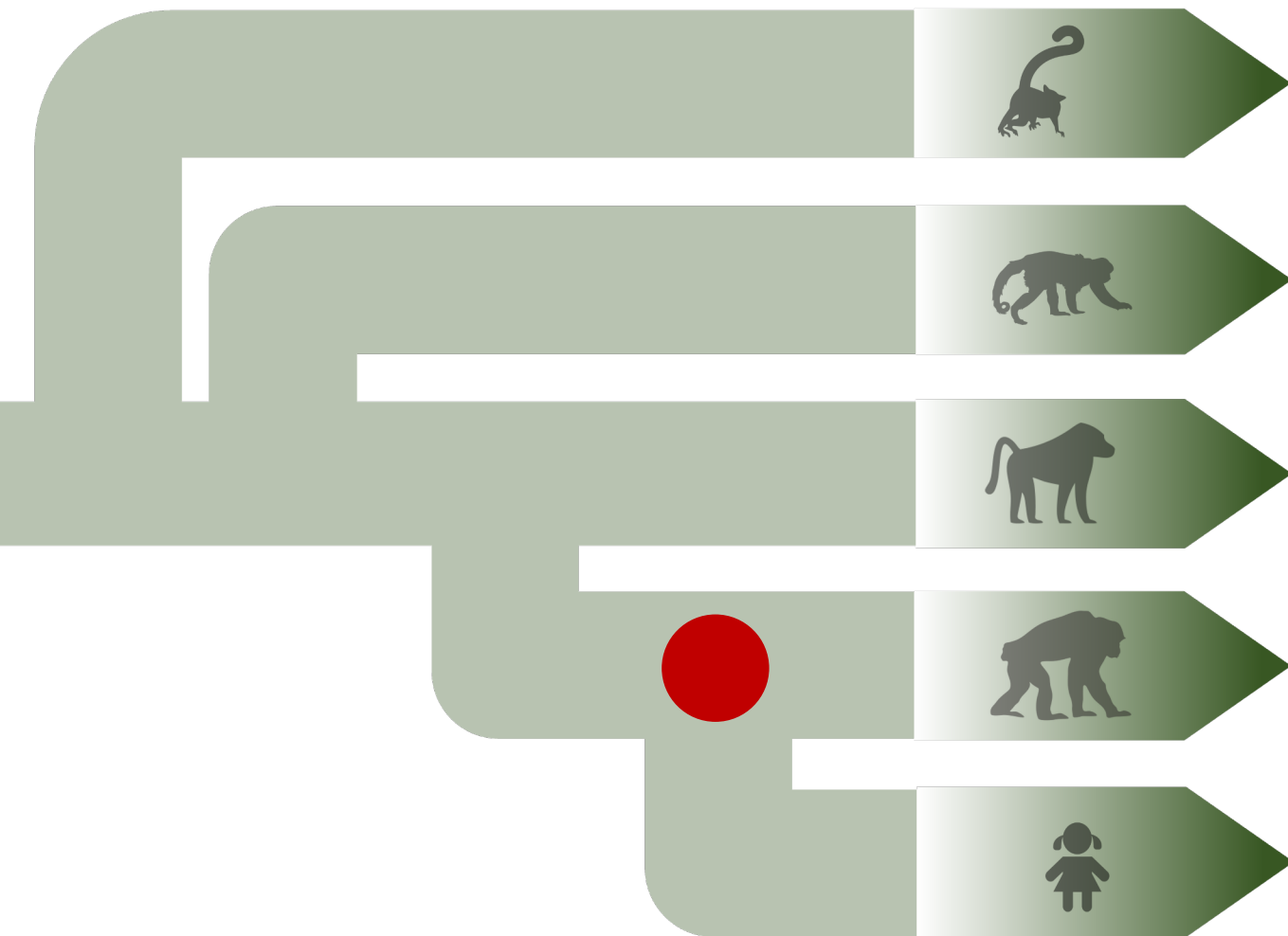
No recursive reasoning



(a) 'Mother-offspring' (left) vs. 'other' (right)



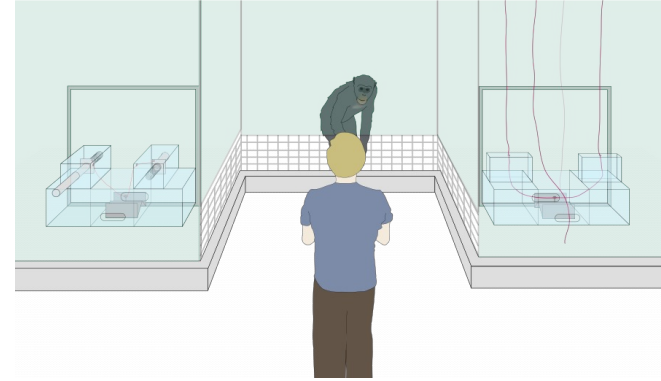
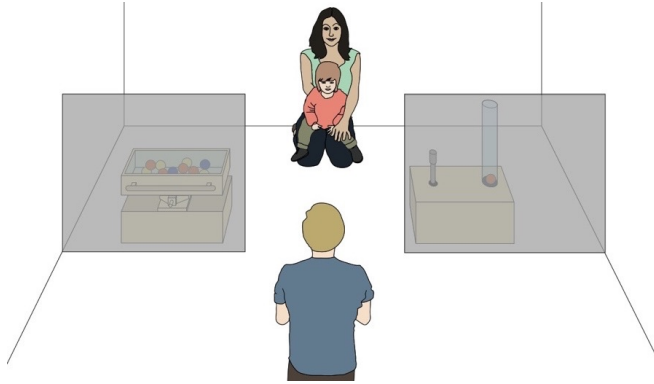
- Which components are necessary (for different species)?
- How are they related to one another?
- Which components are most important?
- What are the semantics of individual signals?
- What changed in the course of human evolution?



Comprehension of ambiguous signals

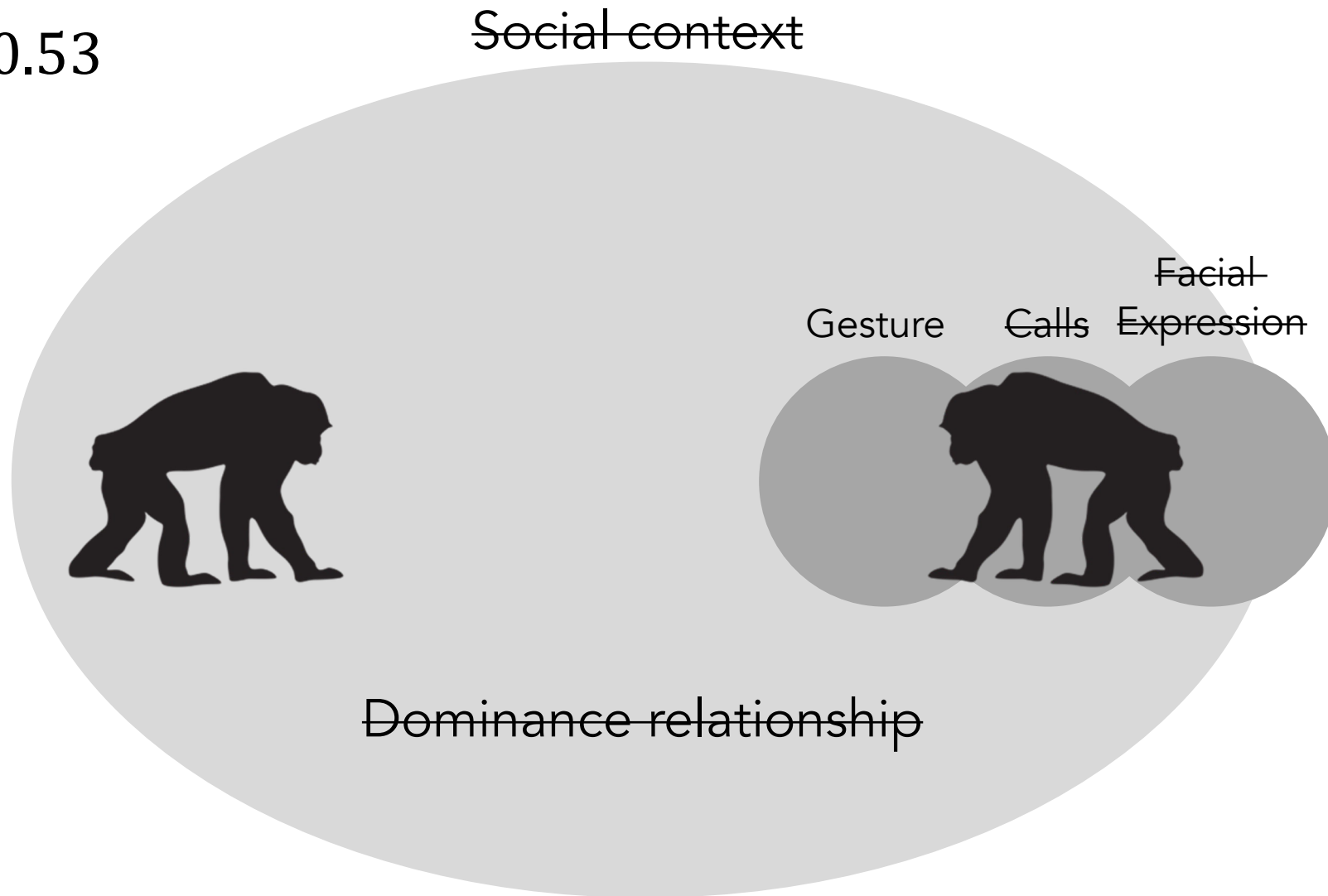


Comprehension of ambiguous signals



Bohn, Kordt, Braun, Call, & Tomasello, 2020
Bohn, Call, & Tomasello, 2019
Margiotoudi, Allritz, Bohn & Pulvermüller, 2019
Margiotoudi, et al., 2022
Bohn, Call, & Tomasello, 2016

👉 $\theta_{gp} = 0.53$



Pragmatic social reasoning as an amplifier of weak meanings?

Semantics of gesture

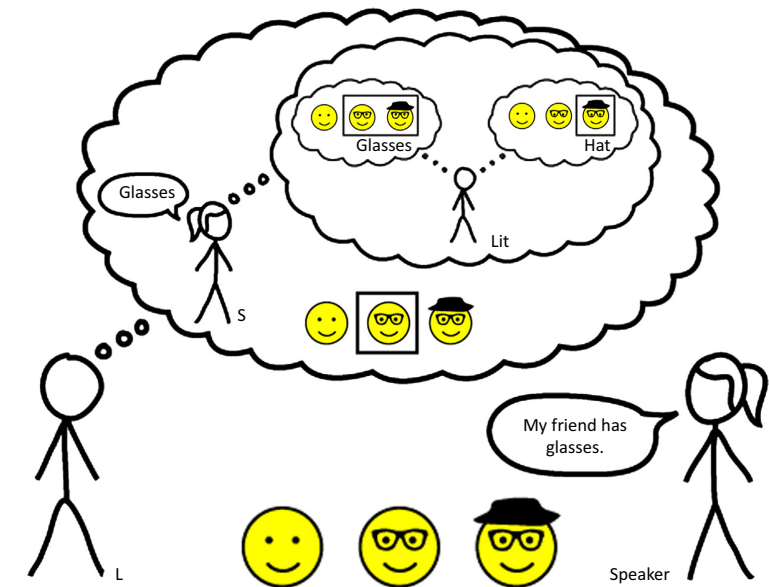
"Literal" s $P_S(u|i) = \mathcal{L}(g, i | \theta_g)$

$$P_R(i|u) \propto P_S(u|i)P(i)$$

	L	R
 <i>point left</i>	0.53 	0.47 
 <i>point right</i>	0.47 	0.53 

Pragmatic (cooperative) sender

$$P_R(i|u) \propto P_S(u|i)P(i)$$



Rational Speech Act Framework

"Literal" receiver

"Pragmatic" S $P_{S_1}(u|i) \propto P_{R_0}(i|u)^\alpha$

$$P_{R_1}(i|u) \propto P_{S_1}(u|i)P(i)$$

"Literal" receiver

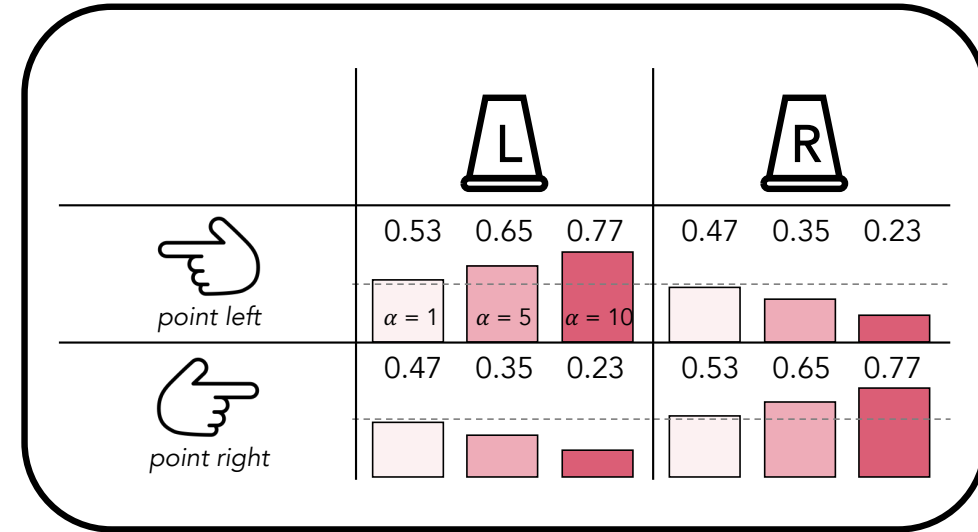
"Pragmatic" S $P_{S_1}(u|i) \propto P_{R_0}(i|u) \alpha$ $\leftarrow$ Informativeness of S

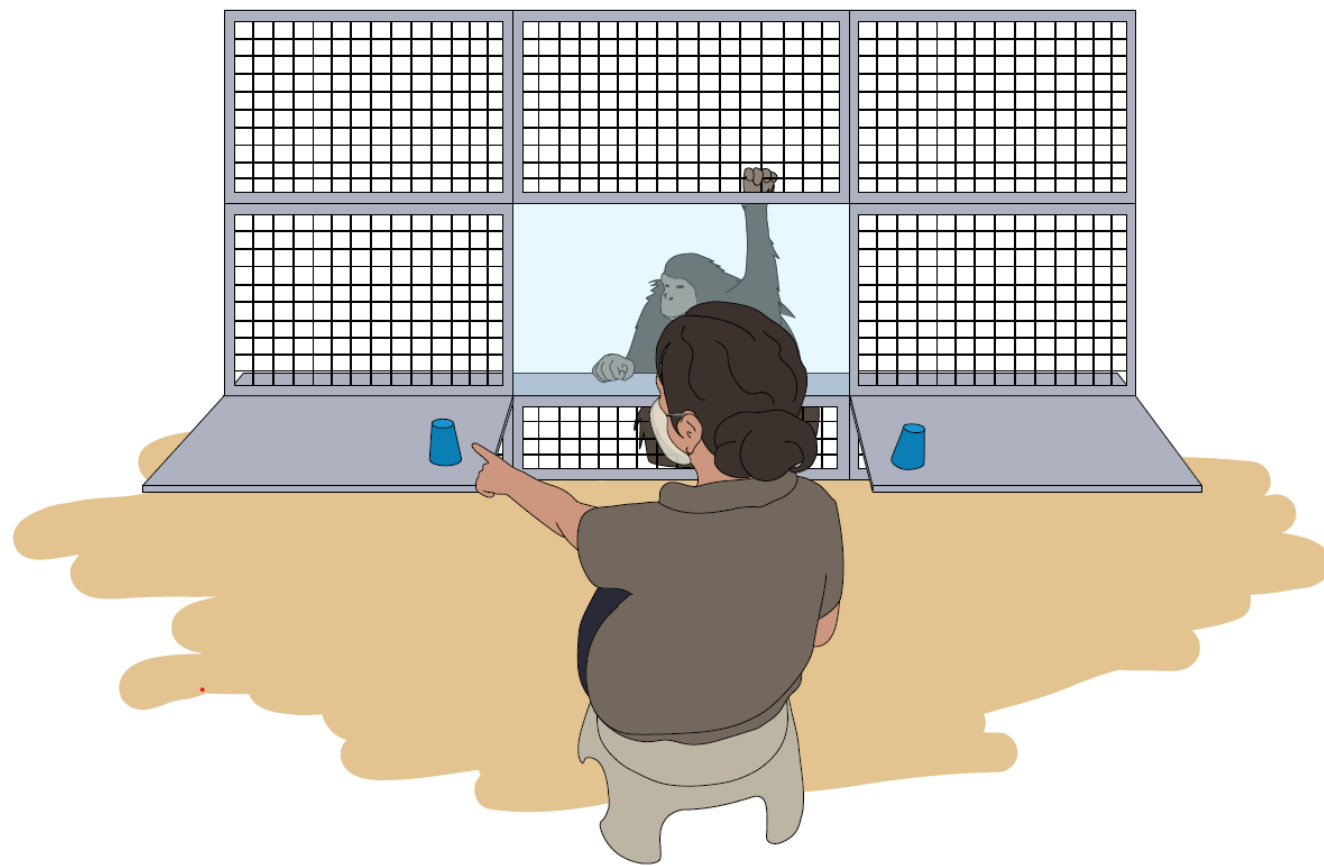
$$P_{R_1}(i|u) \propto P_{S_1}(u|i) P(i)$$

"Amplified" semantics of gesture

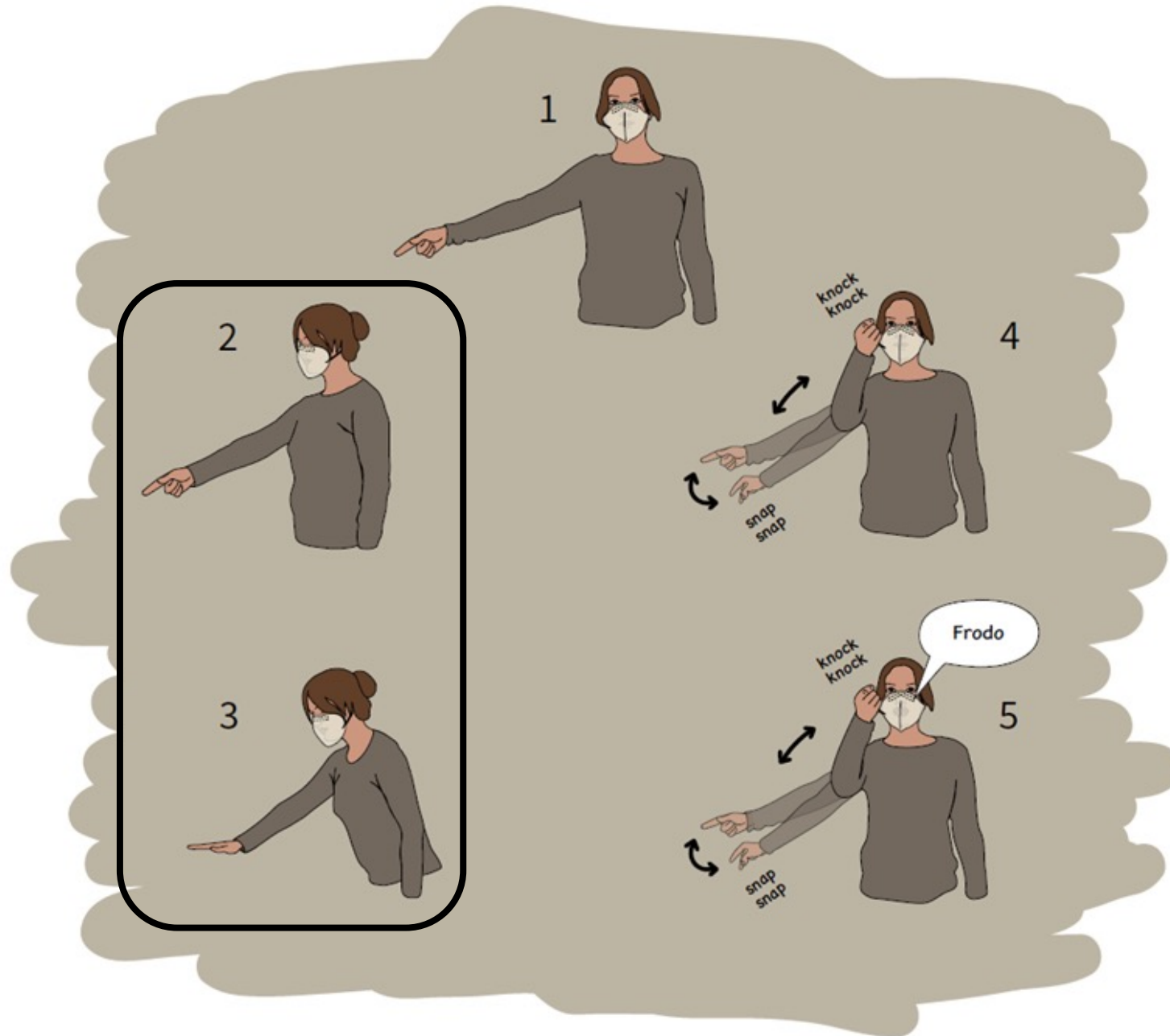
"Pragmatic" S $P_{S_1}(u|i) \propto \mathcal{L}(g, i | \theta_g)^\alpha$

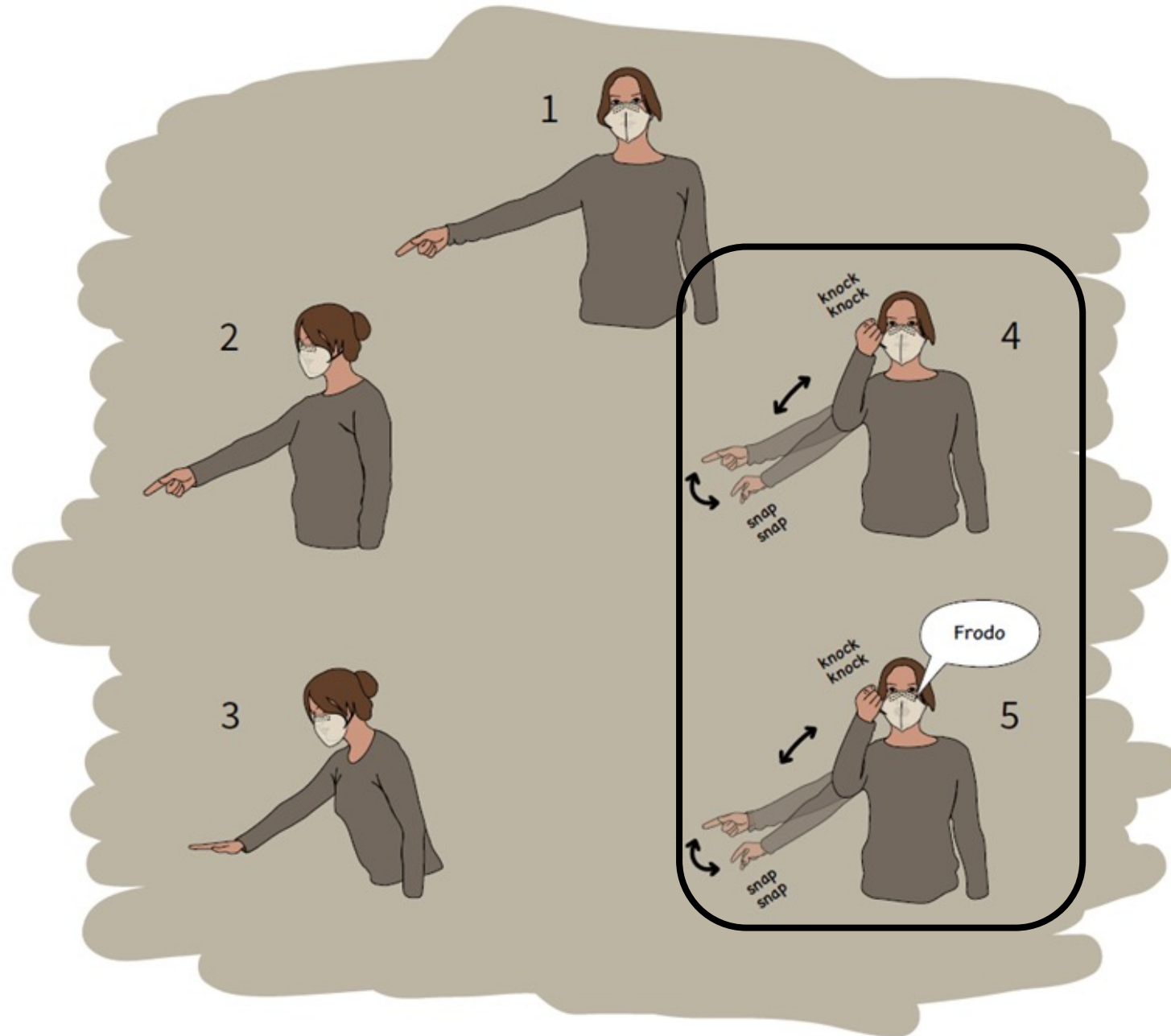
$$P_{R_1}(i|u) \propto P_{S_1}(u|i)P(i)$$



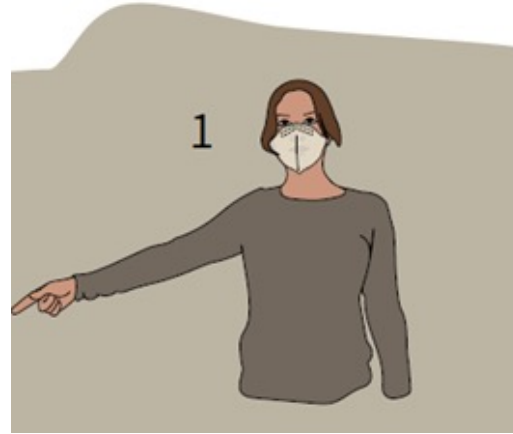


Referential
enrichment



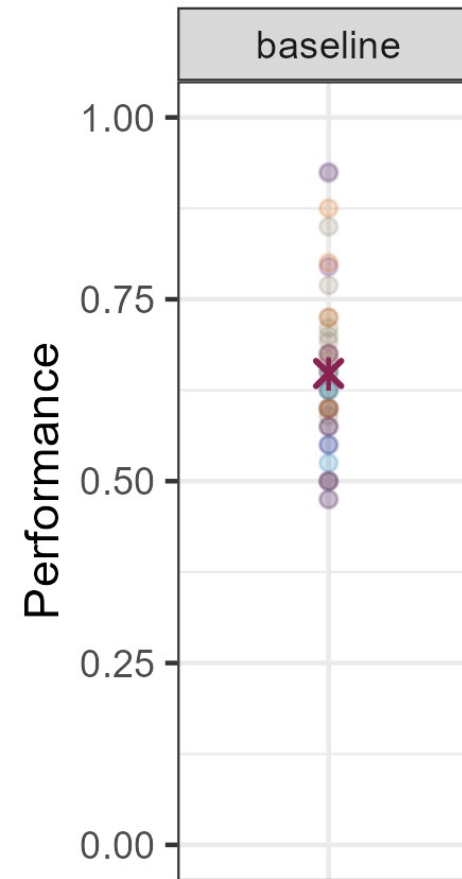


Ostensive
enrichment



Species

- bonobo
- chimpanzee
- gorilla
- orangutan



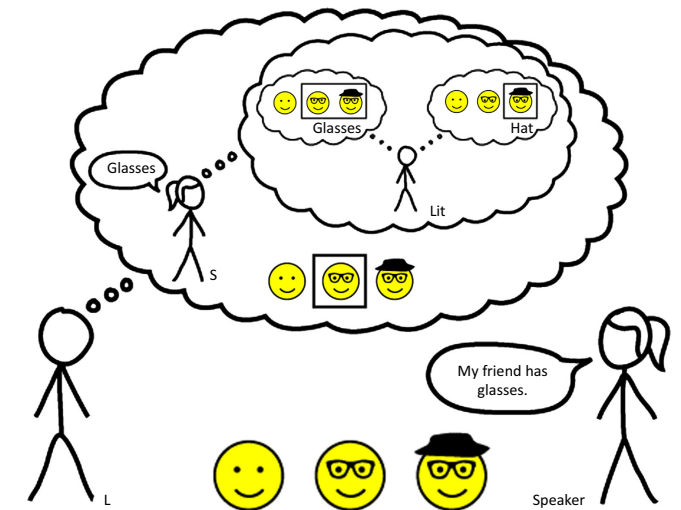
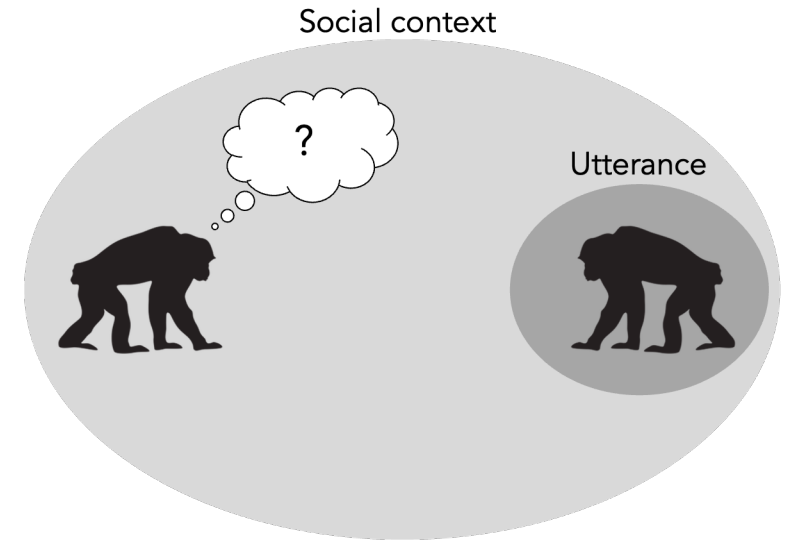
Summary

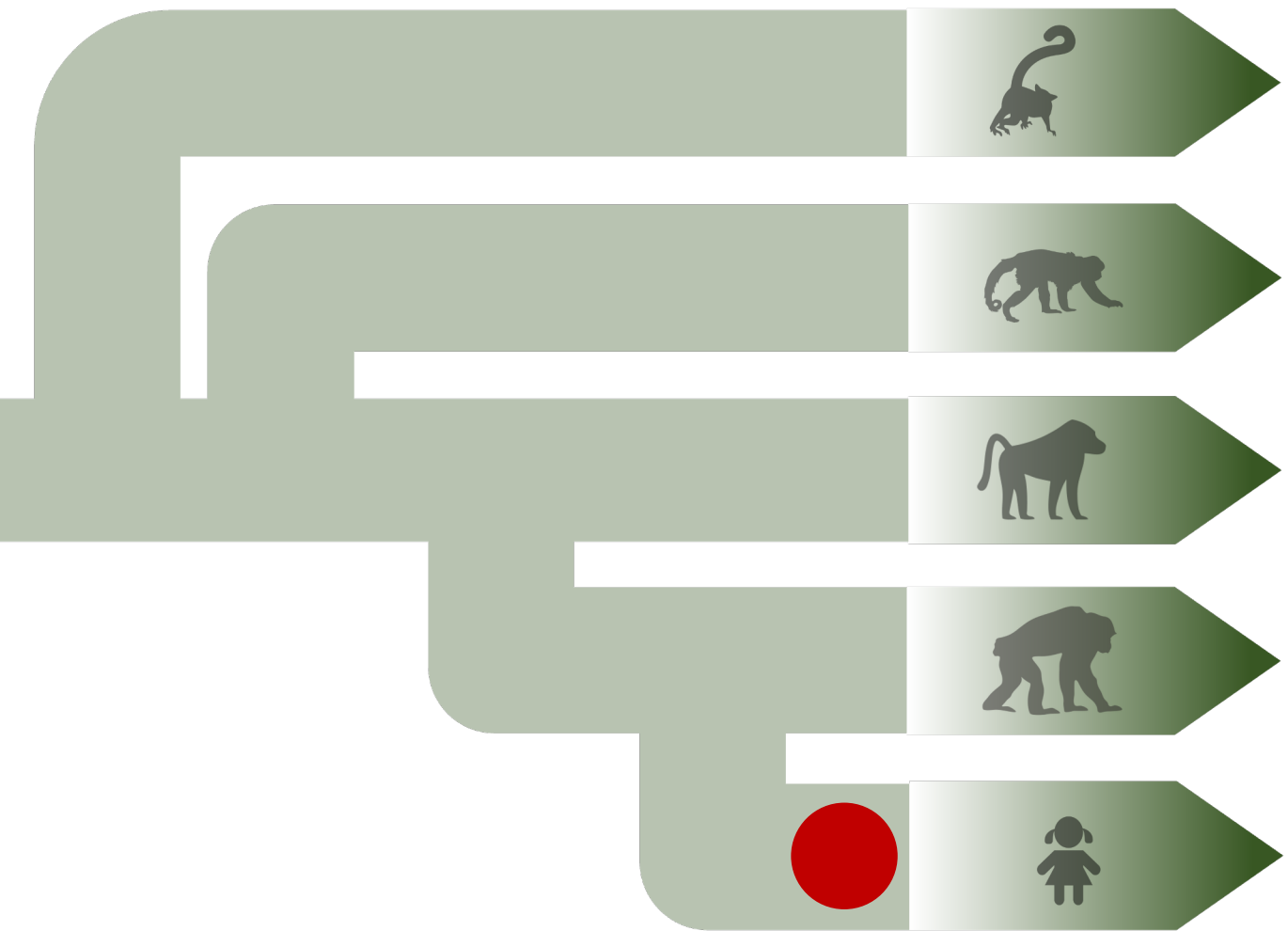
Great ape communication:

- Social inference
- Reasoning based on multiple information sources
- "Rational" (Bayesian) integration of information
- Problems with novel & flexible signals

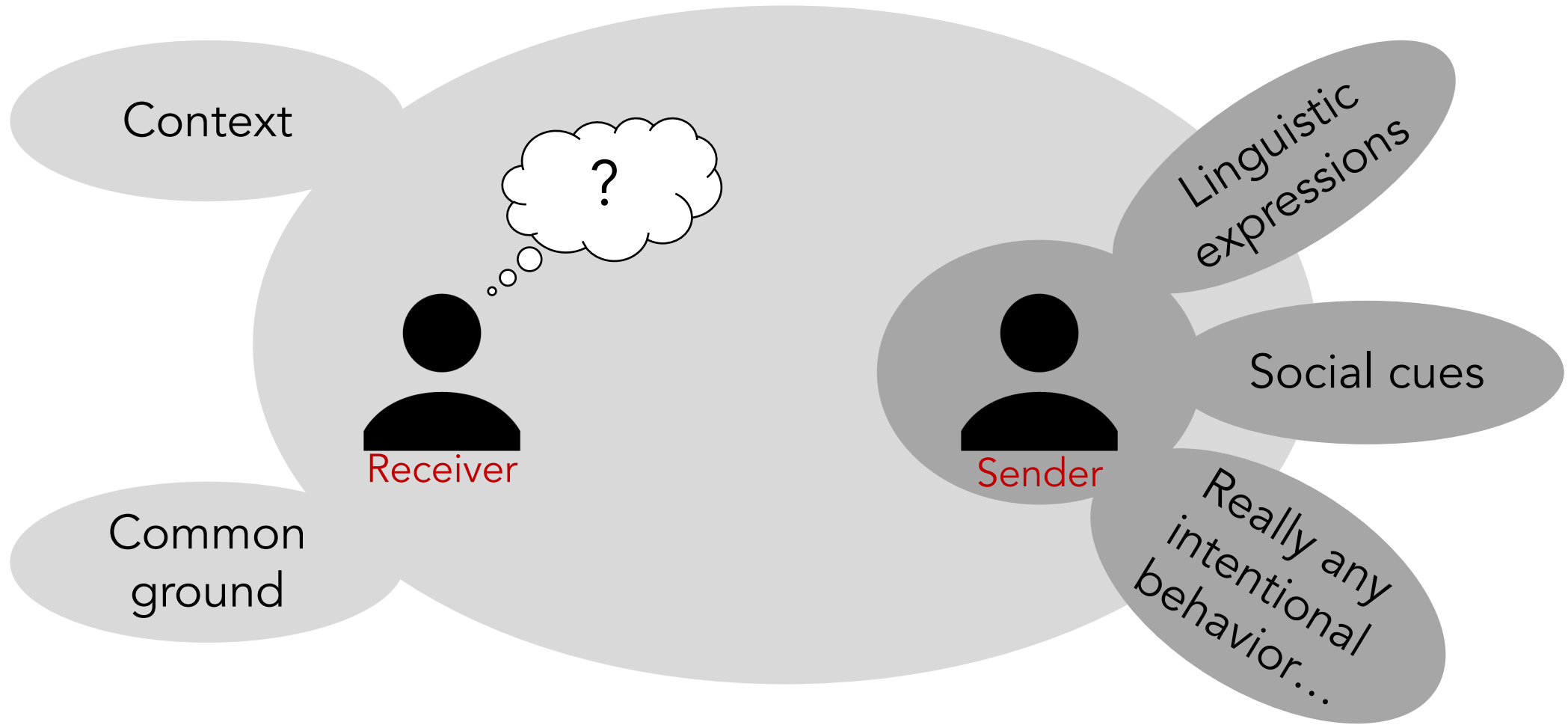
Human communication

- Same basic structure
- Additional social reasoning / pragmatics component





Communication as contextual inference



Grice, 1991
Holler & Levinson, 2014
Scott-Phillips, 2014
Sperber & Wilson, 1986
Tomasello, 2008

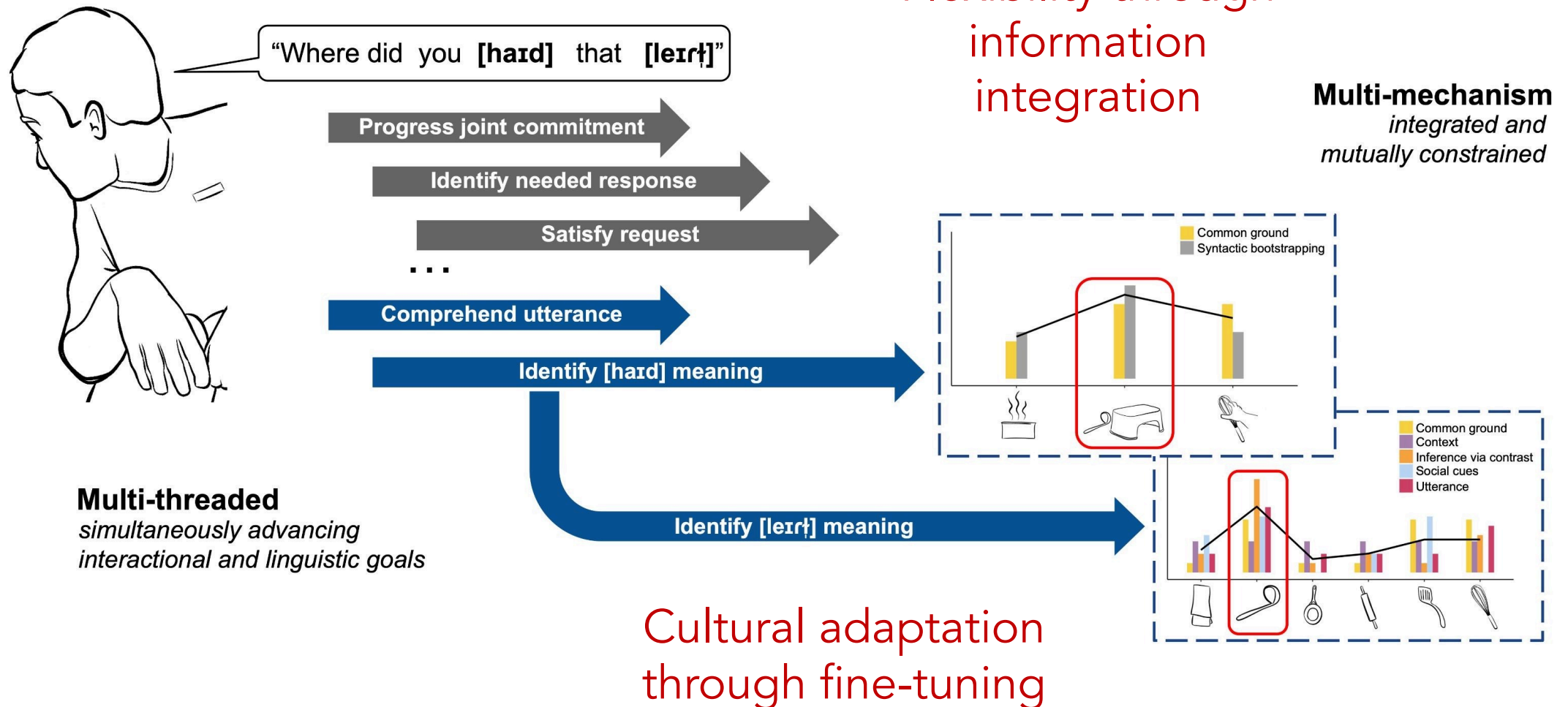
Socio-communicative development

Language learning
is a by-product of
social interaction

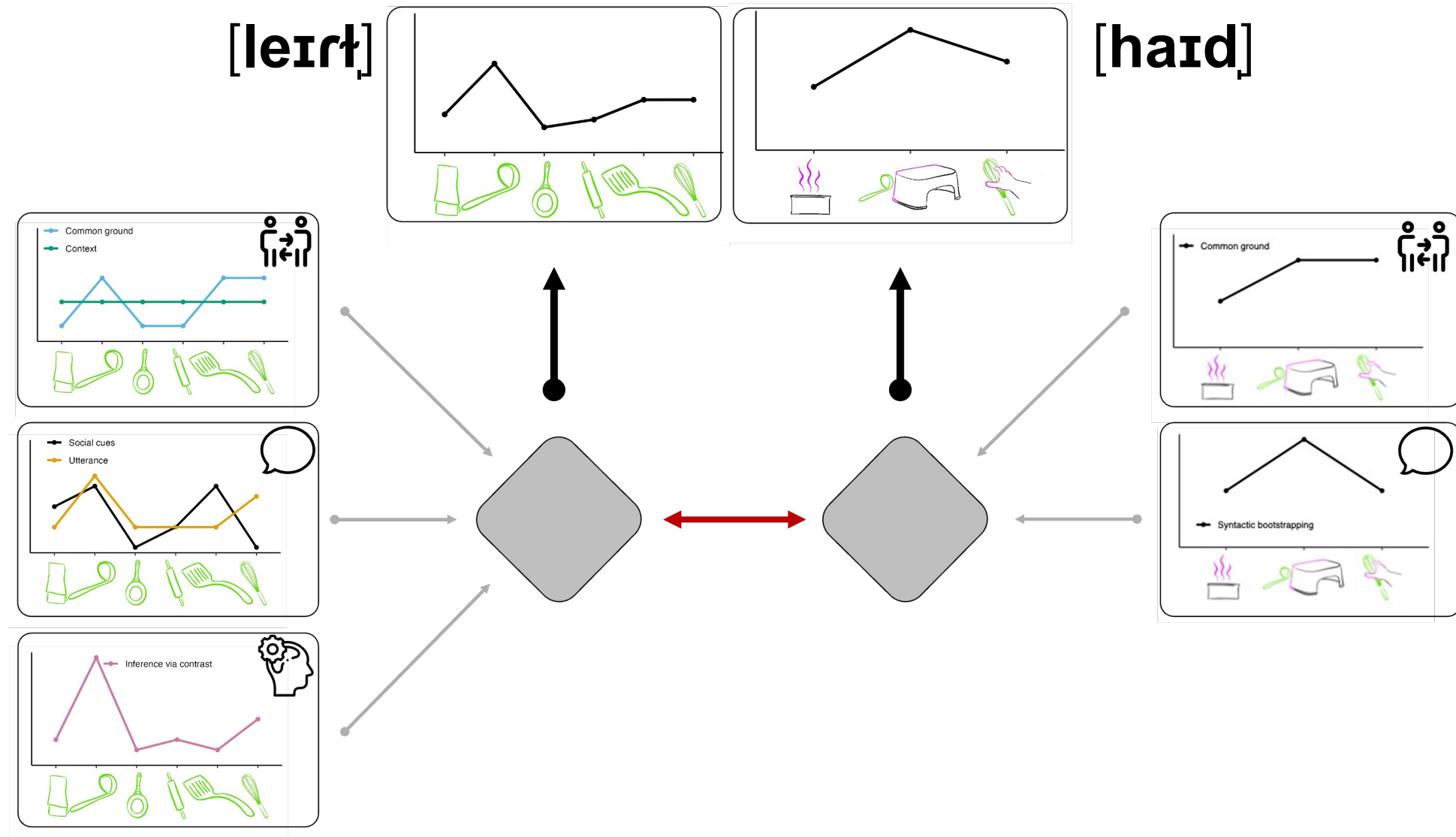


Multiple
information sources
available at any
moment

Socio-communicative development



How are information sources integrated?



How are information sources integrated?

